CURRICULUM VITAE (last update: March 2026)

Dr. Tomáš Hodaň

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Summary: I am a Staff Research Scientist at Meta, working on spatial AI for AR/VR and robotics. My current research focuses on foundation models for 3D vision, particularly zero-shot object detection and pose estimation, as well as language-grounded spatial understanding. I created the BOP benchmark, which became the standard for evaluating object pose estimation, organize R6D workshops, and serve as an Area Chair at major vision conferences. Prior to joining Meta in 2020, I received my PhD in computer vision under the supervision of Prof. Jiri Matas at the Czech Technical University in Prague, during which I interned at Microsoft Research and Google.

Work experience

Meta – Staff Research Scientist (November 2020 – present)
Redmond (USA): 2020 – 2023, Zurich (Switzerland): 2023 – present
Working on spatial AI and deep learning for AR/VR and robotics

Google – Research Intern (November 2018 – May 2019)
Munich (Germany), working with Stefan Hinterstoisser

Microsoft Research – Research Intern (June – August 2018)
Redmond (USA), working with Vibhav Vineet, Sudipta Sinha and Brian Guenter

FP7 EU projects SRS and DARWIN – Computer Vision Researcher (2011 – 2015)
Brno University of Technology, Czech Technical University in Prague
Research and development of household and assembly robots

Education

PhD degree in computer vision (2020, defended in 2021)
Czech Technical University in Prague, Faculty of Electrical Engineering
Doctoral thesis: *Pose Estimation of Specific Rigid Objects*
Supervisor: Prof. Jiří Matas, reviewers: Prof. Vincent Lepetit, Prof. Markus Vincze, Dr. Slobodan Ilic

Master's degree (with honours) in computer science (2013)
Brno University of Technology, Faculty of Information Technology
Master's thesis: *Specular Reflection Detection and Removal From Image Sequences*
Supervisors: Dr. Robby T. Tan and Prof. Adam Herout

Exchange study programme in game and media technology (2011)
Utrecht University, Faculty of Science

Bachelor's degree in computer science (2010)
Brno University of Technology, Faculty of Information Technology

Awards

EgoVis 2023/2024 Distinguished Paper Award, awarded paper: AssemblyHands from CVPR 2023

Best Paper Award at CV4MR CVPR 2025 Workshop, awarded paper: BOP Challenge 2024

Rector's prize for an outstanding PhD thesis (2021), Czech Technical University in Prague

Dean's prize for an excellent Master's thesis (2013), Brno University of Technology

Selected publications

Google Scholar citations: 3971 (January 2026)

SHOW3D: Capturing Scenes of 3D Hands and Objects in the Wild (CVPR 2026)
Patrick Rim, Kevin Harris, B. Copple, S. Han, Xu Xie, Ivan Shugurov, Sizhe An, He Wen, Alex Wong, Tomáš Hodaň, Kun He

HOT3D: Hand and Object Tracking in 3D from Egocentric Multi-View Videos (CVPR 2025)
Prithviraj Banerjee, Sindi Shkodrani, Pierre Moulon, Shreyas Hampali, Shangchen Han, Fan Zhang, Linguang Zhang, Jade Fountain, Edward Miller, Selen Basol, Richard Newcombe, Robert Wang, Jakob Julian Engel, Tomáš Hodaň

FoundPose: Unseen Object Pose Estimation with Foundation Features (ECCV 2024)
Evin Pınar Örnek, Yann Labbé, Bugra Tekin, Lingni Ma, Cem Keskin, Christian Forster, Tomáš Hodaň

In-Hand 3D Object Scanning from an RGB Sequence (CVPR 2023)
Shreyas Hampali, Tomáš Hodaň, Luan Tran, Lingni Ma, Cem Keskin, Vincent Lepetit

AssemblyHands: Towards Egocentric Activity Understanding via 3D Hand Pose Estimation (CVPR 2023)
Takehiko Ohkawa, Kun He, Fadime Sener, Tomáš Hodaň, Luan Tran, Cem Keskin

Neural Correspondence Field for Object Pose Estimation (ECCV 2022)
Lin Huang, Tomáš Hodaň, Lingni Ma, Linguang Zhang, Luan Tran, Christopher Twigg, Po-Chen Wu, Junsong Yuan, Cem Keskin, Robert Wang

LISA: Learning Implicit Shape and Appearance of Hands (CVPR 2022)
Enric Corona, Tomáš Hodaň, Minh Vo, Francesc Moreno-Noguer, Chris Sweeney, Richard Newcombe, Lingni Ma

EPOS: Estimating 6D Pose of Objects with Symmetries (CVPR 2020)
Tomáš Hodaň, Dániel Baráth, Jiří Matas

Learning Surrogates via Deep Embedding (ECCV 2020)
Yash Patel, Tomáš Hodaň, Jiří Matas

BlenderProc: Reducing the Reality Gap with Photorealistic Rendering (RSS Workshops 2020)
Maximilian Denninger, Martin Sundermeyer, Dominik Winkelbauer, Dmitry Olefir, Tomáš Hodaň, Youssef Zidan, Mohamad Elbadrawy, Markus Knauer, Harinandan Katam, Ahsan Lodhi

Photorealistic Image Synthesis for Object Instance Detection (ICIP 2019)
Tomáš Hodaň, Vibhav Vineet, Ran Gal, Emanuel Shalev, Jon Hanzelka, Treb Connell, Pedro Urbina, Sudipta N. Sinha, Brian Guenter

BOP: Benchmark for 6D Object Pose Estimation (ECCV 2018)
Tomáš Hodaň, Frank Michel, Eric Brachmann, Wadim Kehl, Anders Glent Buch, Dirk Kraft, Bertram Drost, Joel Vidal, Stephan Ihrke, Xenophon Zabulis, Caner Sahin, Fabian Manhardt, Federico Tombari, Tae-Kyun Kim, Jiří Matas, Carsten Rother

T-LESS: An RGB-D Dataset for 6D Pose Estimation of Texture-less Objects (WACV 2017)
Tomáš Hodaň, Pavel Haluza, Štěpán Obdržálek, Jiří Matas, Manolis Lourakis, Xenophon Zabulis

Activities

Organized the R6D workshop series on object pose estimation – http://cmp.felk.cvut.cz/sixd
(at ICCV'15, ECCV'16, ICCV'17, ECCV'18, ICCV'19, ECCV'20, ECCV'22, ICCV'23, ECCV'24, ICCV'25)

Organized BOP challenges on object pose estimation – http://bop.felk.cvut.cz

Other co-organized events:
OpenCV Perception Challenge for Bin-Picking (CVPR'25)
PIRA Workshop on Perception for Industrial Robotics Automation (CVPR'25)

Area chair at CVPR'25, CVPR'26, ECCV'26; regular reviewer at top-tier conferences

Teaching – Pattern Recognition and Machine Learning course at the Faculty of Electrical Engineering, Czech Technical University in Prague (2013/14, 2014/15, 2017/18)